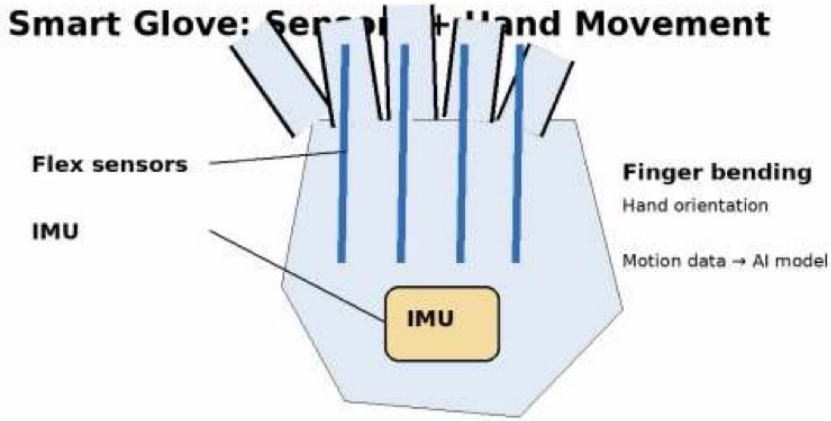
AI-Based Smart Glove for Hand Movement Recognition

Introduction to the technology and its role in rehabilitation

Based on the systematic review by Mohammed & Ali (2026)

Journal of Engineering and Applied Science, 73:246



Why Hand Rehabilitation Matters

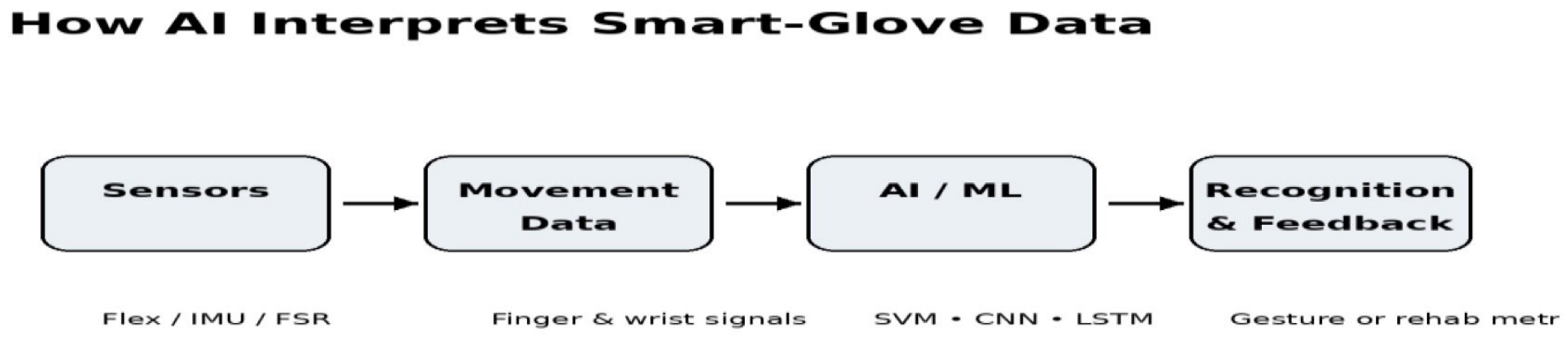
- Stroke and neuromuscular disorders can significantly affect hand motor function.
- Hand impairment may cause weakness, rigidity, poor fine-motor control, and reduced independence.
- Traditional rehabilitation can be time-intensive and usually depends on repeated clinical supervision.

What Is a Smart Glove?

- A wearable glove equipped with sensors that capture hand and finger movements.
- Common sensors include flex sensors, inertial measurement units (IMUs), and force-sensitive resistors (FSRs).
- The collected sensor data can be converted into digital information for gesture recognition and rehabilitation monitoring.

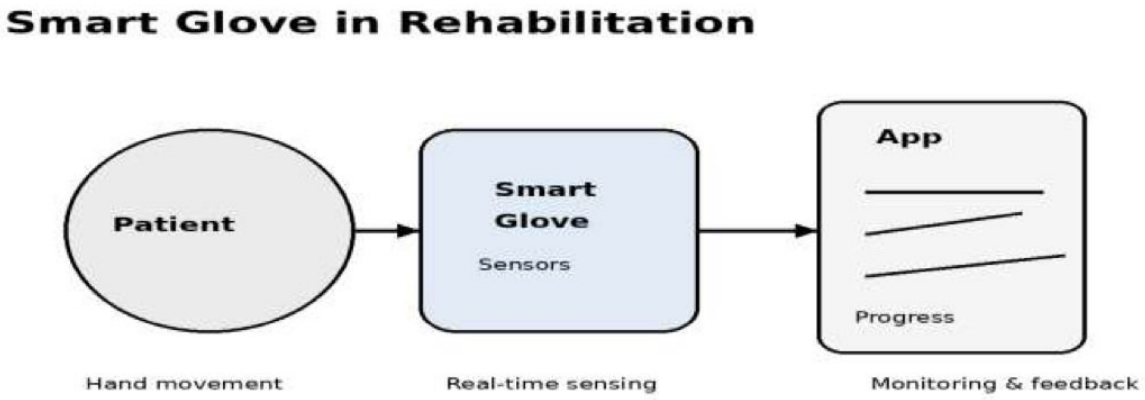
Role of AI in Smart Gloves

- AI and machine learning can classify hand gestures and analyze movement patterns.
- Common approaches include SVM, Random Forest, CNN, LSTM, and other deep-learning models.
- The review reports recognition accuracies generally ranging from 82% to 99% across included studies.
- Multimodal sensing, especially flex + IMU, is widely used to improve motion recognition.



Why This Technology Is Important

- Smart gloves can support real-time monitoring and personalized rehabilitation.
- They can enable quantitative tracking of hand kinematics and movement performance.
- Potential applications include post-stroke rehabilitation, tele-rehabilitation, and home-based monitoring.
- Key challenges remain: sensor drift, limited clinical validation, comfort, power efficiency, and standardized datasets.

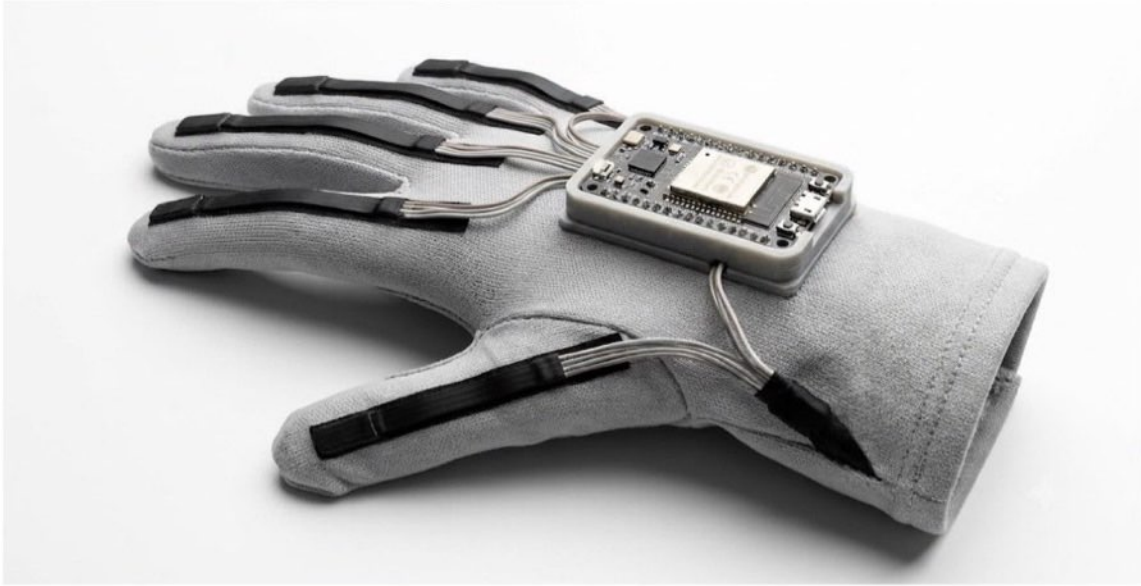


Illustrations created for this presentation

SECTION 3 • HARDWARE & SENSOR ARCHITECTURE

Smart Glove: Sensors & Hardware

Transducing Hand Movement into Digital Data for Rehabilitation



Core Sensors Used in Smart Gloves



1. Resistive Flex Sensors

Placed along the fingers to detect bending and joint angles at the MCP, PIP, and DIP joints.

2. Inertial Measurement Units (IMUs)

Mounted on the back of the hand or wrist to track 3D orientation, tilt, pitch, and roll.

3. Force-Sensitive Resistors (FSRs)

Positioned at the fingertips to evaluate grasping ability, object contact, and pinch strength.

Sensor Fusion: Combining Modalities



Why Single Sensors Are Not Enough:

Flex sensors cannot detect hand orientation, while IMUs alone cannot distinguish individual finger movements.

The 72% Multimodal Standard:

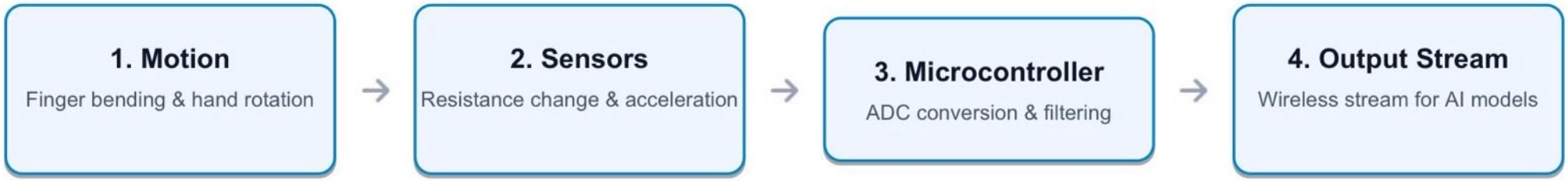
72% of the systematic review studies combine flex sensors and IMUs to achieve accurate, drift-compensated tracking.

Embedded Microcontroller Hub:

Compact low-power chips (ESP32 or Arduino) sample both sensor streams synchronously in real time.

From Hand Motion to Digital Data

How physical hand movements are converted into clean sensor streams



On-Board Signal Conditioning:

Raw electrical signals pass through voltage dividers and digital low-pass filters to eliminate trembling and sensor noise.

Real-Time Wireless Transmission:

Bluetooth Low Energy (BLE) or Wi-Fi sends frames with under 15 ms latency, enabling instant feedback during therapy.

Hardware Challenges & Practical Limitations



1. Sensor Drift & Degradation

Resistive flex and force sensors degrade mechanically over thousands of bends, requiring regular recalibration.

2. Glove Fit & Donning Difficulties

Accurate sensing requires a snug fit, but stroke patients with hand spasticity or stiffness struggle to put tight gloves on.

3. Wiring & Battery Trade-Offs

Multiple sensors add bulky wiring, while keeping the glove lightweight limits battery runtime during at-home sessions.

The Need for Effective Hand Rehabilitation



When Movement Becomes A Challenge

- 93.8 million people were living with stroke globally in 2021.
- 11.9 million new strokes occurred in 2021.
- Stroke can cause long-term motor impairments, affecting movement and coordination.
- Rehabilitation is essential for recovering functional ability and independence.
- Hand and upper-limb recovery is particularly important for everyday activities.

Source: WHO, Stroke Fact Sheet, 2025

Impact on Everyday Life

Loss of hand movement can affect:

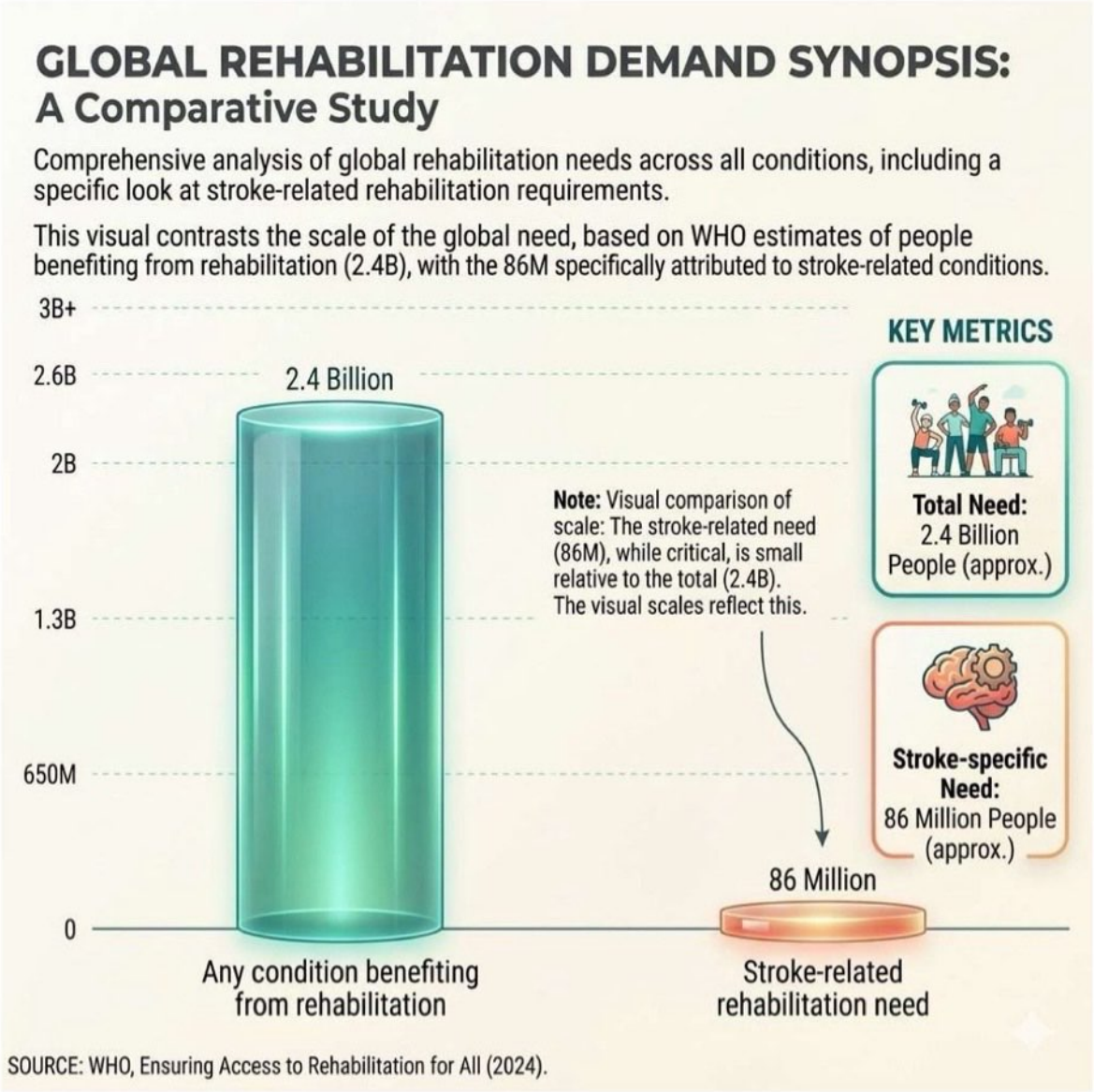
- Writing and using a keyboard
- Eating and handling objects
- Dressing and personal care
- Using phones and other devices
- Performing everyday tasks independently



The Rehabilitation Challenge

- 86 million people globally need rehabilitation because of stroke.
- Rehabilitation often requires repeated exercises over an extended period.
- Patients may need to continue exercises outside supervised clinical sessions.
- Physiotherapists cannot continuously observe every exercise performed at home.

This raises questions like: Are the exercises being performed correctly? Are enough repetitions being completed?
Is the patient's movement improving?



The Monitoring Gap

Conventional approach:

- Patient performs exercise → Physiotherapist observes → Periodic assessment

What is missing?

- Continuous movement data
- Objective repetition counting
- Quantitative measurement of movement
- Immediate feedback on incorrect movements
- Data that can be reviewed over time

Technology opportunity:

- Hand movement → Sensors → Digital data → Analysis → Feedback

The Problem We Aim To Solve

Smart Glove for Physiotherapy Tracking

We aim to develop a wearable system that can:

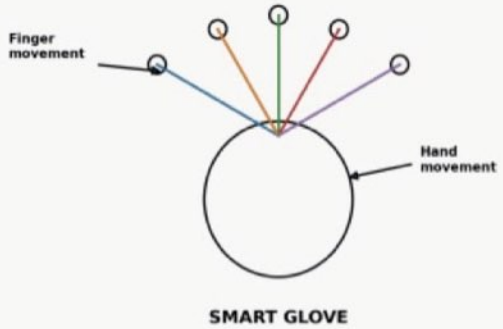
- **Track** finger and hand movements
- **Count** physiotherapy repetitions
- **Measure** movement characteristics
- **Identify** correct and incorrect movements using ML
- **Provide** measurable feedback on rehabilitation performance

The Problem Statement

Patients undergoing hand rehabilitation often need to perform repetitive exercises both in the clinic and at home. However, without continuous supervision, it can be difficult to know whether the exercises are being performed correctly, whether enough repetitions are completed, or whether movement is improving over time. Our problem is to develop an accessible system that can objectively track hand movements, count repetitions, assess movement correctness, and provide feedback to support effective physiotherapy.

SMART GLOVE:
FROM HAND MOVEMENT TO USEFUL FEEDBACK

A simple view of how wearable sensing can support hand rehabilitation



What Makes the Glove “Smart”?

A normal glove covers the hand.

A smart glove adds sensors that can notice movement.

So the glove can:

- Detect finger bending
- Detect hand movement and orientation
- Record the movement as digital data
- Send the data for analysis

Simple idea: the glove acts like a “movement recorder” for the hand.

Instead of only asking “Did they exercise?”



we can ask “How did they move?”

A Simple Exercise Example

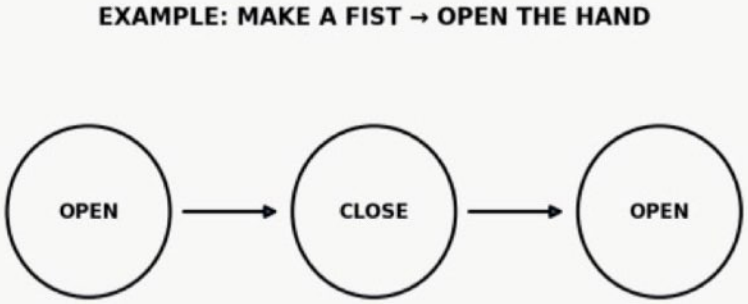
Consider one basic exercise: making a fist and opening the hand.

During the exercise, the fingers change their bending position.

The smart glove records that change.

This can help us measure:

- How many times the exercise was repeated
- How long each movement took
- Whether the movement pattern was consistent



The glove detects the change in finger bending.

What Do the Sensors Actually Do?

FLEX SENSOR

Tells us how much a finger bends.

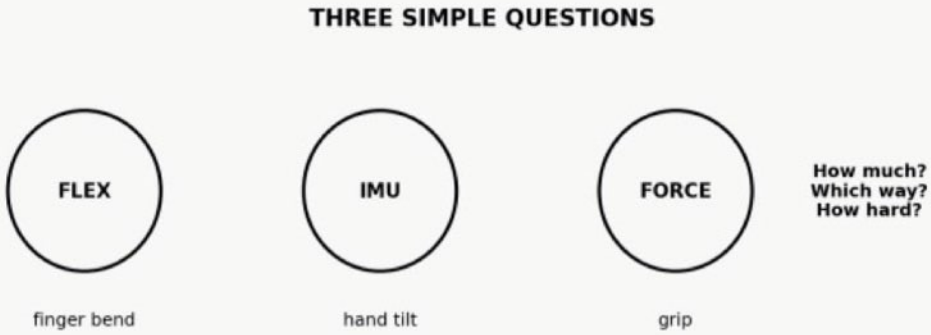
IMU

Tells us how the hand is moving or tilted.

FORCE SENSOR

Can be added when the exercise involves gripping or pressing.

Each sensor gives one piece of information. Together, they give a clearer picture of the hand movement.



How Does AI Enter the Picture?

The glove first collects sensor readings.

Then:

1. The signal is cleaned
2. Useful movement information is extracted
3. An ML model looks for patterns
4. The system gives a result

For example, the model can learn the difference between different exercise movements.

The 2026 systematic review reports that smart-glove research commonly uses models such as SVM, Random Forest, CNN and LSTM.

The computer turns movement into an easy-to-read result.



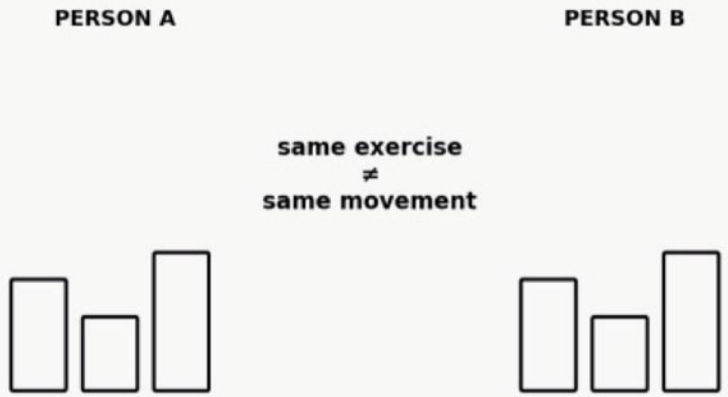
Why Personalization Matters

People do not move their hands in exactly the same way.

One user may bend a finger more than another user while doing the same exercise.

Therefore, the system can use calibration to understand the user's normal movement range.

Goal: measure improvement in the individual user instead of simply comparing different people.



Calibration helps the glove adapt to the user.

From Prototype to Rehabilitation Tool

STEP 1 — Build the glove

Choose sensors and connect them to a microcontroller.

STEP 2 — Record exercises

Collect movement data from selected exercises.

STEP 3 — Train the model

Teach the system to recognize movement patterns.

STEP 4 — Test

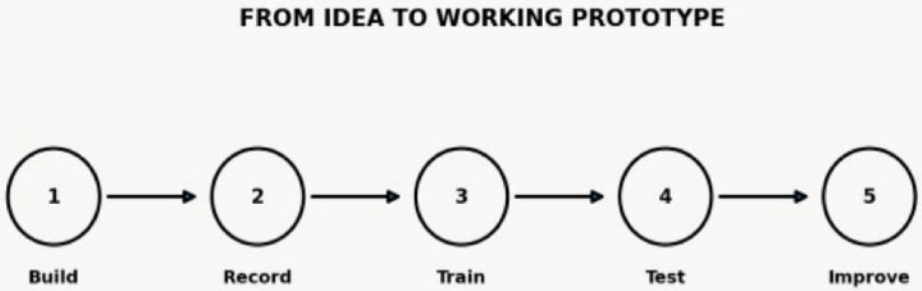
Check repetition counting and movement recognition.

STEP 5 — Improve

Work on comfort, calibration, battery life and reliability.

FINAL IDEA

Movement → Sensors → Data → ML → Feedback



REHABILITATION TECHNOLOGY • CAPSTONE PROJECT PROPOSAL

Smart Glove for Physiotherapy Tracking

A sensor-equipped glove that tracks hand and finger movement during rehabilitation exercises, using machine learning to score correctness and count repetitions.

Hardware

Flex + IMU sensors

Software

ML-based rep scoring

Domain

Rehab medicine

01 Introduction

Why Physiotherapy Needs a Digital Eye

Physiotherapy for hand and finger injuries depends heavily on patients performing prescribed exercises correctly and consistently at home, between clinical visits. In practice, therapists have almost no way to verify how well — or how often — a patient actually performs these movements once they leave the clinic.

This gap leads to poor exercise adherence, incorrect movement patterns going uncorrected, and slower or incomplete recovery. A wearable that can sense motion and give objective, real-time feedback directly addresses this problem.

THE PROBLEM, IN NUMBERS

50%+

of home-exercise programs are performed with reduced or incorrect adherence

No

objective way for therapists to verify form once the patient leaves the clinic

Manual

rep counting and self-reporting remain the clinical default today

02 Objectives

Project Objectives

1 Design a wearable sensor glove

Instrument a glove/arm-band with flex sensors and an IMU to accurately capture finger and hand joint motion.

2 Digitize rehab exercises

Translate standard physiotherapy hand exercises into measurable motion signals suitable for automated analysis.

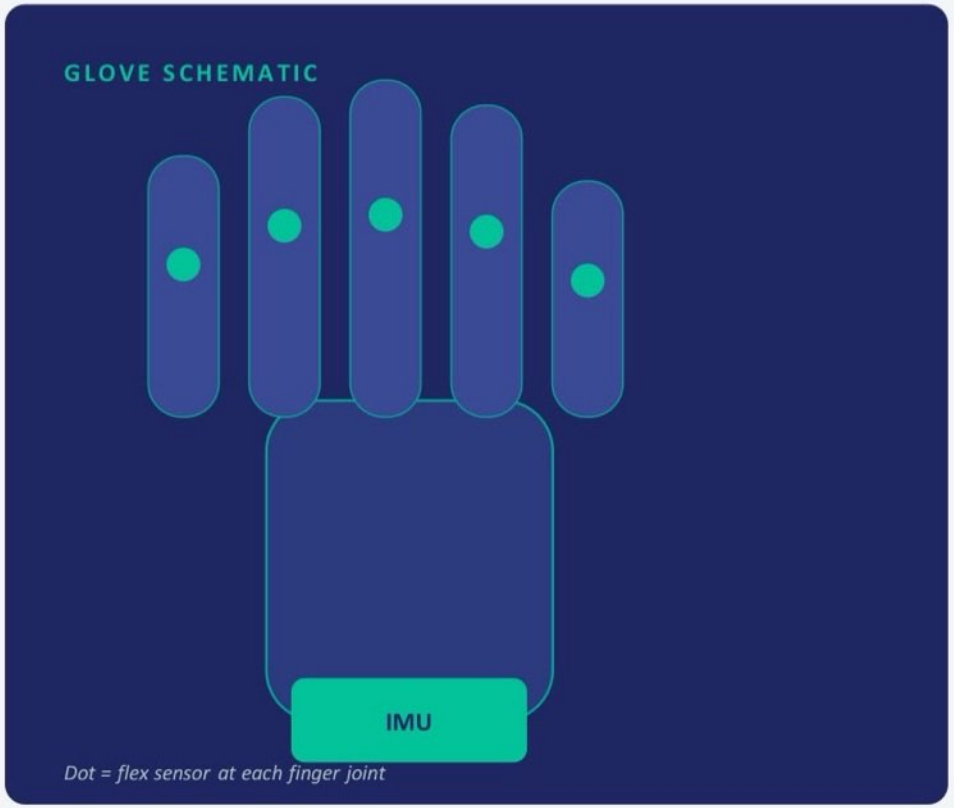
3 Build an ML scoring model

Train a lightweight model to score movement correctness and automatically count valid repetitions.

4 Deliver real-time feedback

Provide the patient and therapist with a simple, immediate readout of performance and progress.

Hardware Design



Flex sensors

One per finger joint to capture bend angle during each exercise repetition.

IMU (accelerometer + gyroscope)

Mounted at the wrist to capture overall hand orientation and motion trajectory.

Microcontroller (e.g., ESP32)

Reads sensor data and streams it over Bluetooth/Wi-Fi to the companion app.

Rechargeable battery + enclosure

Keeps the glove lightweight, wireless, and comfortable for repeated clinical or home use.

03 Methodology

Data & ML Pipeline

1

Sensor Streaming

Flex + IMU readings sampled continuously during each exercise session and sent to the app.

2

Preprocessing

Signals are filtered, normalized, and segmented into individual repetition windows.

3

Feature Extraction

Joint-angle ranges, motion smoothness, and timing features computed per repetition.

4

ML Classification

A trained model scores movement correctness and flags reps as valid, partial, or incorrect.

5

Feedback & Logging

Rep count and correctness score shown to the patient and logged for the therapist to review.

Model approach: A supervised classifier (e.g., Random Forest or a lightweight neural network) trained on labeled repetitions performed correctly vs. incorrectly under therapist supervision.

04 Scope

Project Scope

IN SCOPE — This Project

- Glove prototype covering finger flexion and wrist orientation tracking
- Support for a defined set of standard hand/finger physiotherapy exercises
- ML model for rep counting and correctness scoring on those exercises
- Companion app showing real-time feedback and session history
- Bench and small-scale user testing to validate sensor accuracy

FUTURE SCOPE — Beyond This Phase

- Extending sensing to full-arm and multi-joint rehabilitation protocols
- Clinical trials in partnership with physiotherapy departments
- Personalized ML models that adapt to individual patient baselines
- Direct dashboard integration for remote therapist monitoring
- Miniaturization and regulatory pathway toward a certified medical device

EXPECTED OUTCOME

From guesswork to measurable recovery

A low-cost, wearable system that turns physiotherapy exercises into objective, trackable data — helping patients recover with confidence and giving therapists visibility they don't have today.

Thank You